

Cross-family Morphology Evaluation Summary

Project: Cross-family morphological transfer evaluation

Seed: 42

Languages: aze, tur, hun, fin

Number of evaluated train-test pairs: 16

Average monolingual exact match: 0.9666

Average cross-lingual exact match: 0.3052

Main observations:

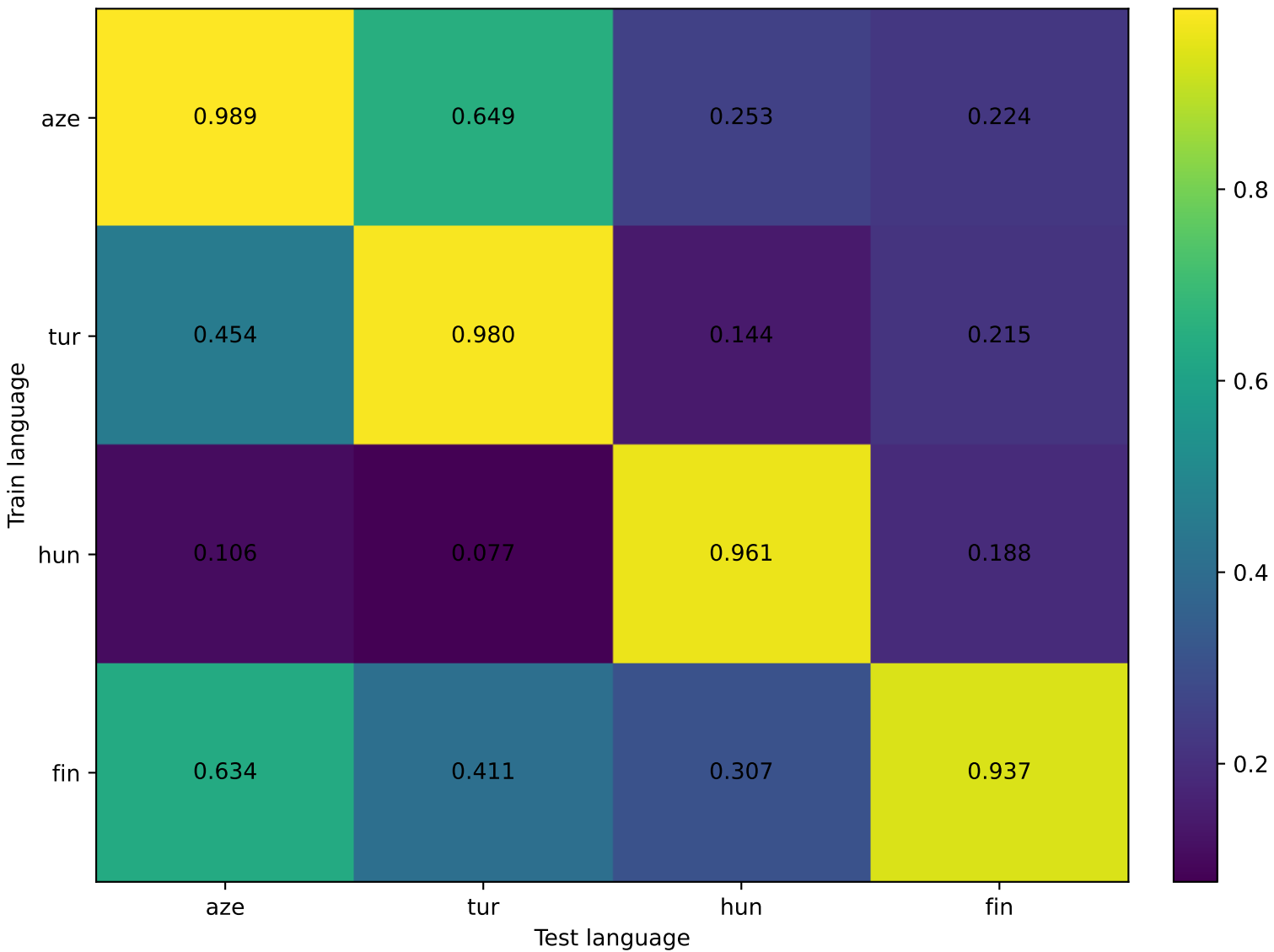
- Monolingual performance is high for all four languages.
- Azerbaijani and Turkish show the strongest cross-lingual transfer.
- Hungarian is the weakest source language for transfer to other languages.
- Finnish transfers relatively well to Azerbaijani and Hungarian compared with other non-diagonal settings.

Exact Match Matrix

	aze	tur	hun	fin
aze	0.9886	0.6488	0.2531	0.2244
tur	0.4543	0.9796	0.1439	0.2150
hun	0.1057	0.0766	0.9614	0.1880
fin	0.6343	0.4107	0.3073	0.9369

Rows indicate training language. Columns indicate test language.

Exact Match Heatmap

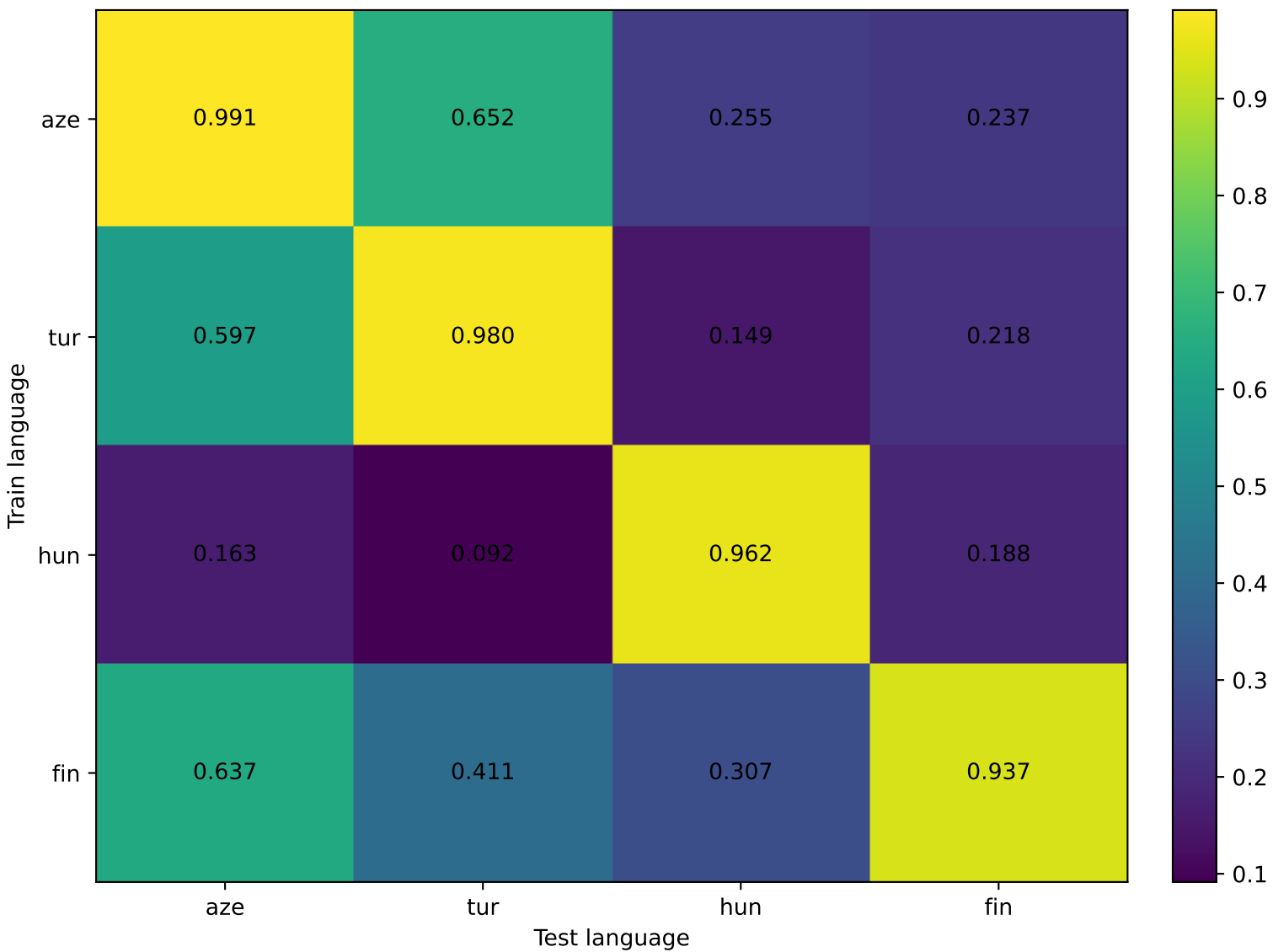


Stem Accuracy Matrix

	aze	tur	hun	fin
aze	0.9914	0.6521	0.2550	0.2366
tur	0.5971	0.9796	0.1491	0.2182
hun	0.1629	0.0915	0.9619	0.1884
fin	0.6371	0.4111	0.3074	0.9369

Rows indicate training language. Columns indicate test language.

Stem Accuracy Heatmap

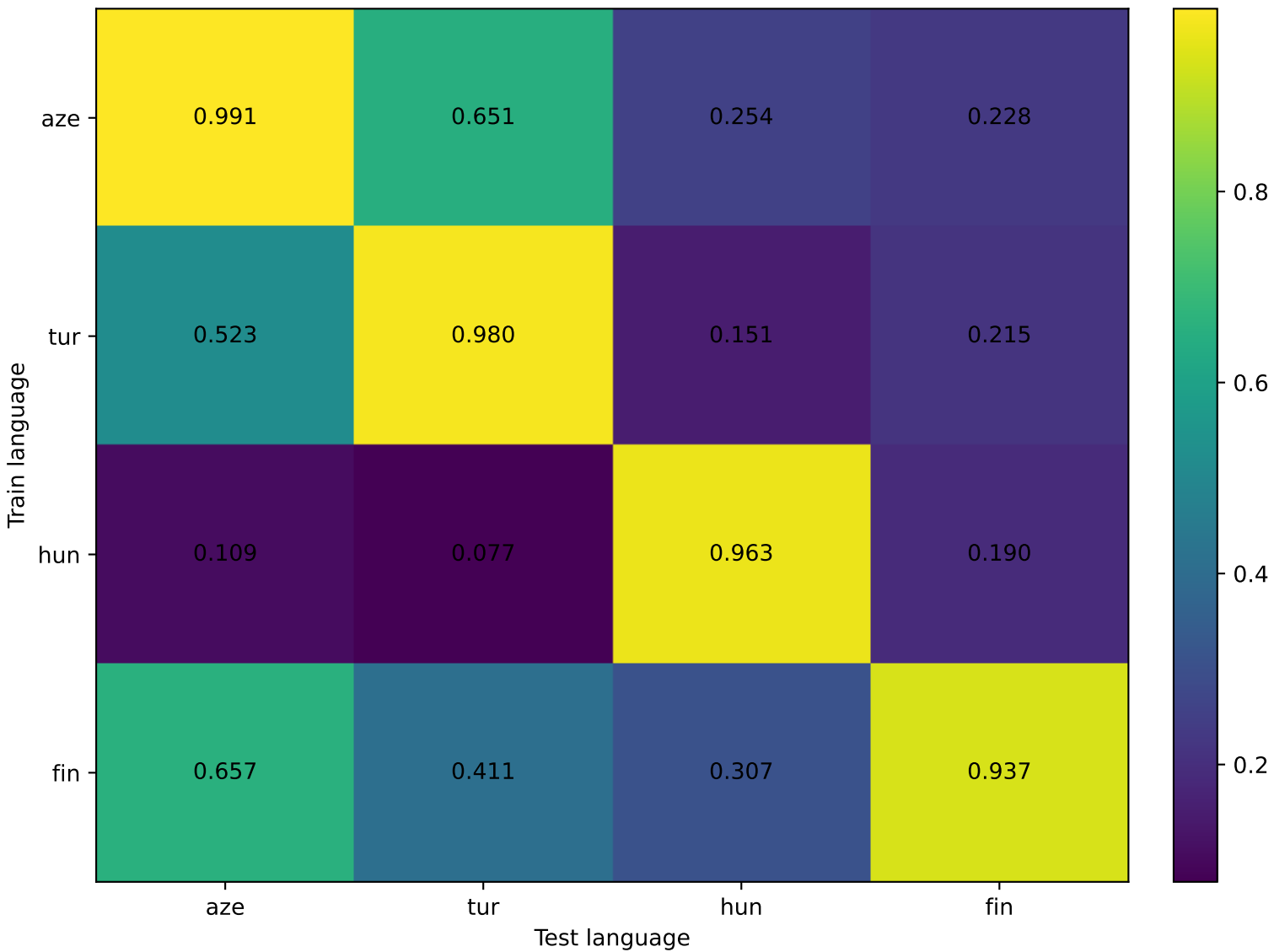


Suffix Accuracy Matrix

	aze	tur	hun	fin
aze	0.9914	0.6512	0.2538	0.2280
tur	0.5229	0.9797	0.1506	0.2154
hun	0.1086	0.0771	0.9632	0.1896
fin	0.6571	0.4107	0.3073	0.9369

Rows indicate training language. Columns indicate test language.

Suffix Accuracy Heatmap



Cross-lingual Transfer Ranking

Best cross-lingual transfer pairs by exact match:

aze -> tur: Exact Match = 0.6488, Stem = 0.6521, Suffix = 0.6512

fin -> aze: Exact Match = 0.6343, Stem = 0.6371, Suffix = 0.6571

tur -> aze: Exact Match = 0.4543, Stem = 0.5971, Suffix = 0.5229

fin -> tur: Exact Match = 0.4107, Stem = 0.4111, Suffix = 0.4107

fin -> hun: Exact Match = 0.3073, Stem = 0.3074, Suffix = 0.3073

Worst cross-lingual transfer pairs by exact match:

hun -> tur: Exact Match = 0.0766, Stem = 0.0915, Suffix = 0.0771

hun -> aze: Exact Match = 0.1057, Stem = 0.1629, Suffix = 0.1086

tur -> hun: Exact Match = 0.1439, Stem = 0.1491, Suffix = 0.1506

hun -> fin: Exact Match = 0.1880, Stem = 0.1884, Suffix = 0.1896

tur -> fin: Exact Match = 0.2150, Stem = 0.2182, Suffix = 0.2154

Full Evaluation Results

train_lang	test_lang	total_examples	exact_match	stem_accuracy	suffix_accuracy
aze	aze	350	0.9886	0.9914	0.9914
aze	tur	12824	0.6488	0.6521	0.6512
aze	hun	46079	0.2531	0.2550	0.2538
aze	fin	2553	0.2244	0.2366	0.2280
tur	aze	350	0.4543	0.5971	0.5229
tur	tur	12824	0.9796	0.9796	0.9797
tur	hun	46079	0.1439	0.1491	0.1506
tur	fin	2553	0.2150	0.2182	0.2154
hun	aze	350	0.1057	0.1629	0.1086
hun	tur	12824	0.0766	0.0915	0.0771
hun	hun	46079	0.9614	0.9619	0.9632
hun	fin	2553	0.1880	0.1884	0.1896
fin	aze	350	0.6343	0.6371	0.6571
fin	tur	12824	0.4107	0.4111	0.4107
fin	hun	46079	0.3073	0.3074	0.3073
fin	fin	2553	0.9369	0.9369	0.9369